

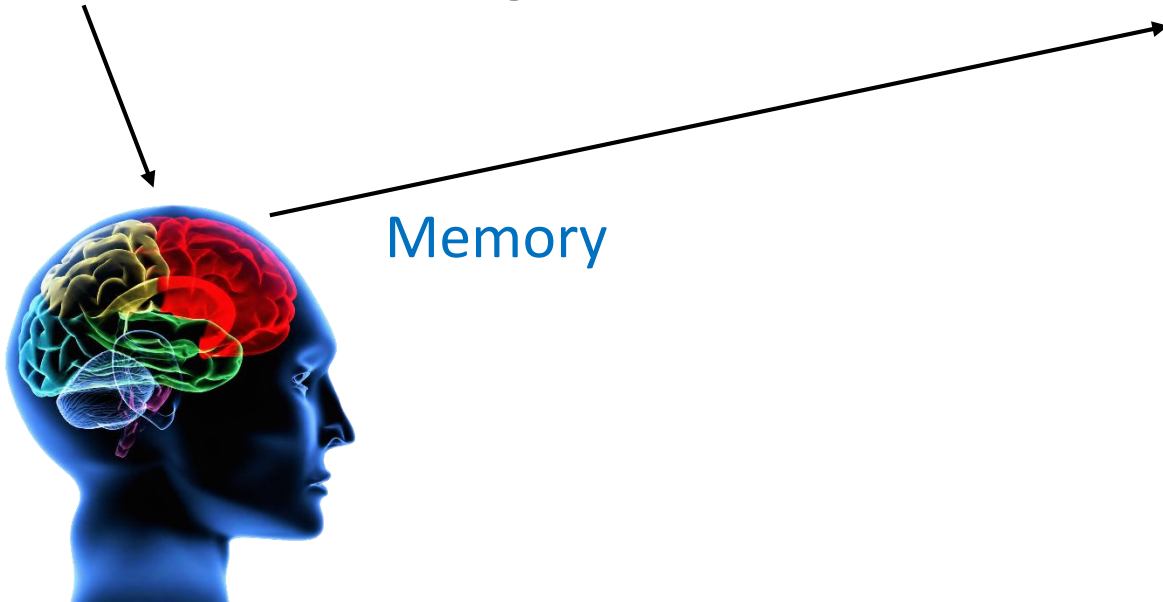
Chapter 2: Memory & Learning

Digital Education (CS-411)

September 15, 2026

To understand how people learn, we need to ‘understand’ the constraints of our memory system.

Learning **produces** knowledge and skills that can be **used later**.



Experiment 1

Name as many cities as possible in 30 seconds.

Experiment 2

Memorize the following list of cities during 30 seconds.

1. Manchester
2. Brussels
3. Lille
4. Athens
5. Lisbon
6. Amsterdam
7. Paris
8. Washington
9. Camberra
10. Tunis
11. Montreal
12. Tokyo
13. Oslo
14. Berlin
15. Bratislava
16. Tashkent
17. Beijing
18. Calcutta
19. Marseille
20. Basel

Sensory Memory
(< 1 s)

The list is see / hear / ...

Working Memory
(< 1 min)

The list I memorized

Long Term Memory
($\pm$ unlimited)

All the cities I know

Cognitive Psychology

To understand how people learn, we need to ‘understand’
the constraints of our memory system.

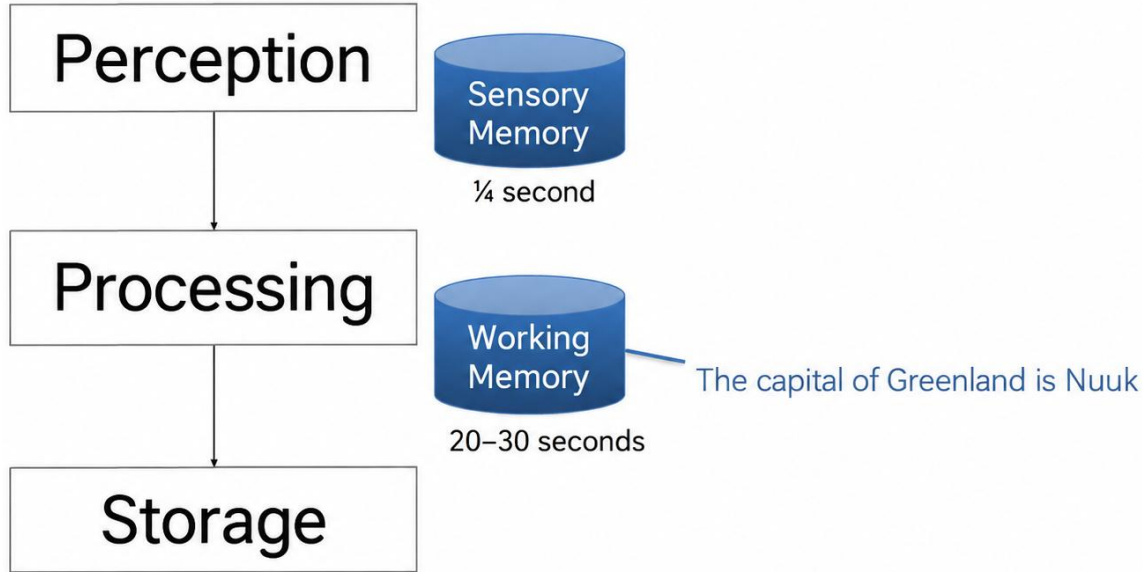
But learning is much more than memorizing.



NUUK

An information processing model

Information



Knowledge

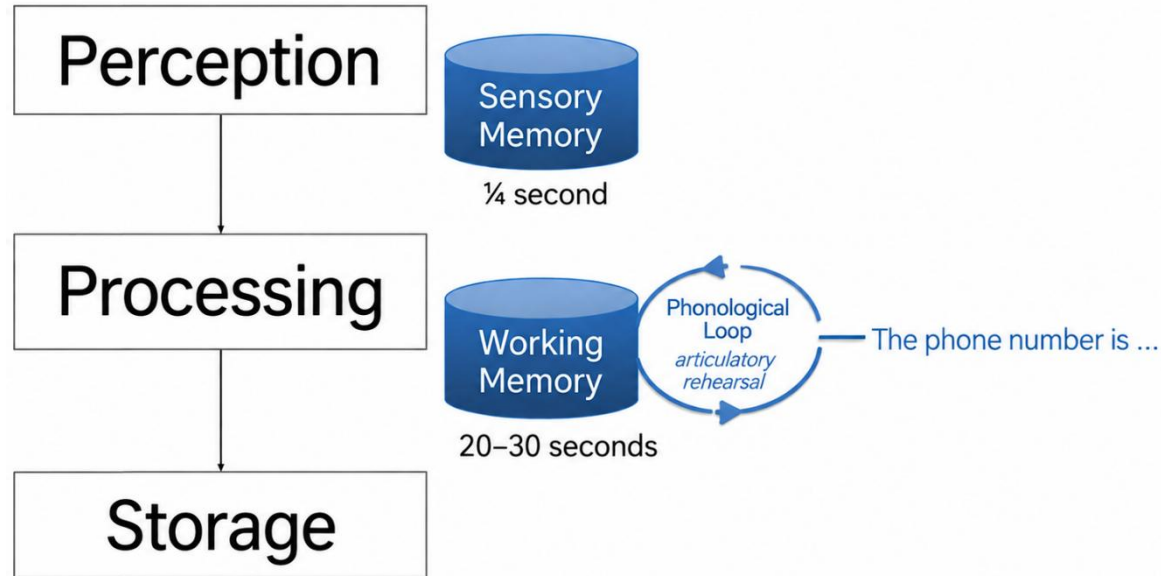
My phone number is: 078 725 36 92



My phone number is:

An information processing model

Information



Knowledge

My car plate is: VD 312 864

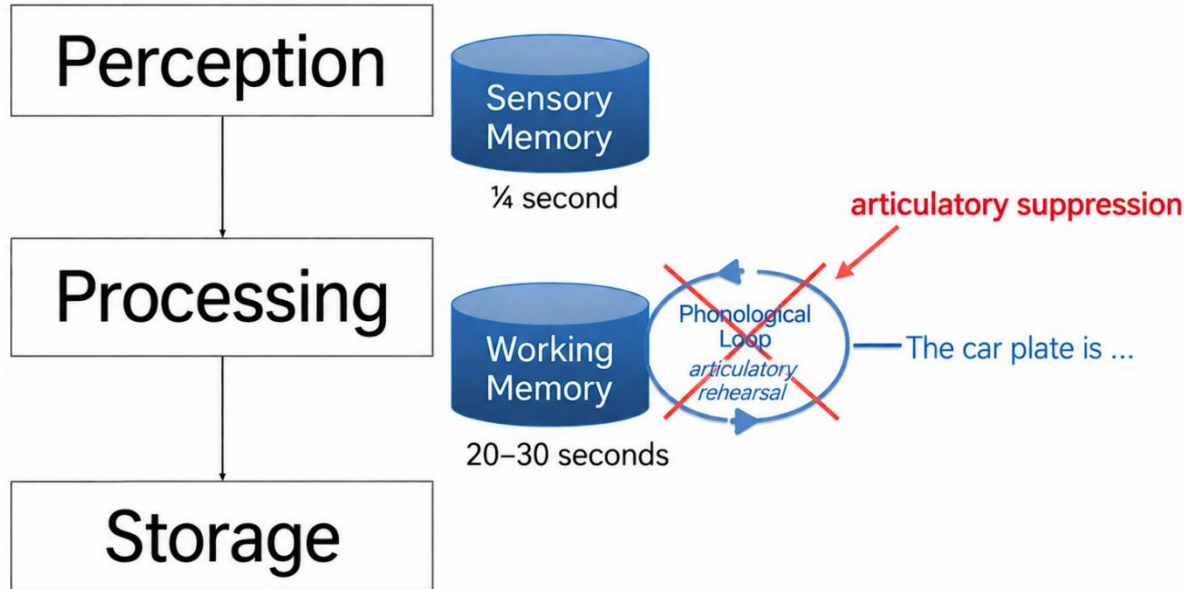
Read Aloud

For a long time, Nuuk was occupied by the Dorset culture around the former settlement of Kangeq but they disappeared from the Nuuk district before AD 1000. The Nuuk area was later inhabited by Viking explorers in the 10th century (Western Settlement), and shortly thereafter by Inuit peoples.

My car plate is:

An information processing model

Information



Knowledge

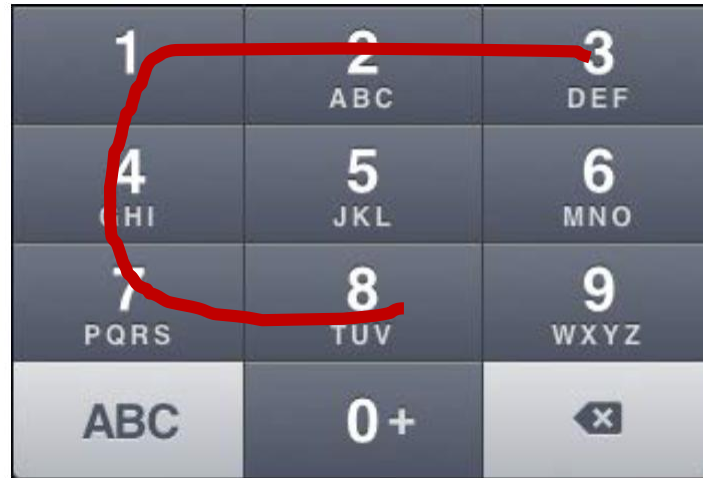
My passcode is: 321478



Read Aloud

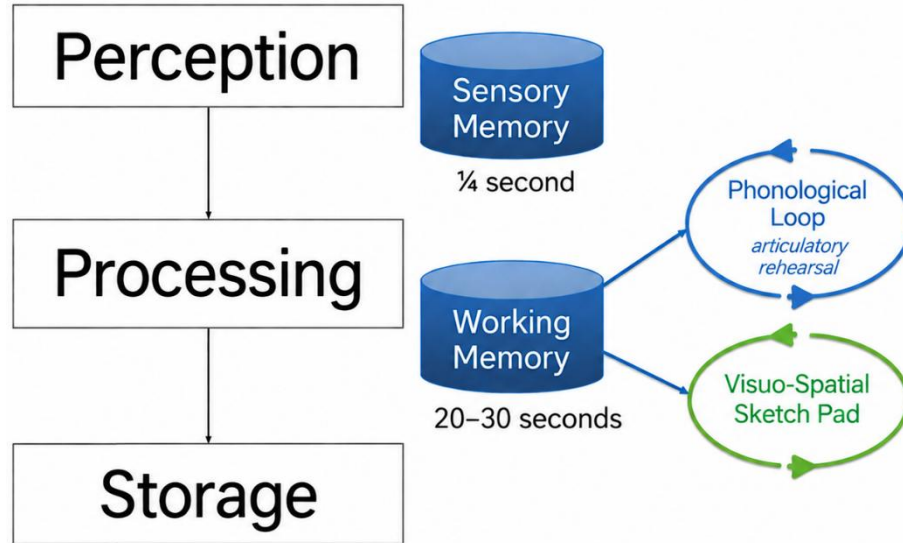
For a long time, Nuuk was occupied by the Dorset culture around the former settlement of Kangeq but they disappeared from the Nuuk district before AD 1000. The Nuuk area was later inhabited by Viking explorers in the 10th century (Western Settlement), and shortly thereafter by Inuit peoples.

What is my passcode?



An information processing model

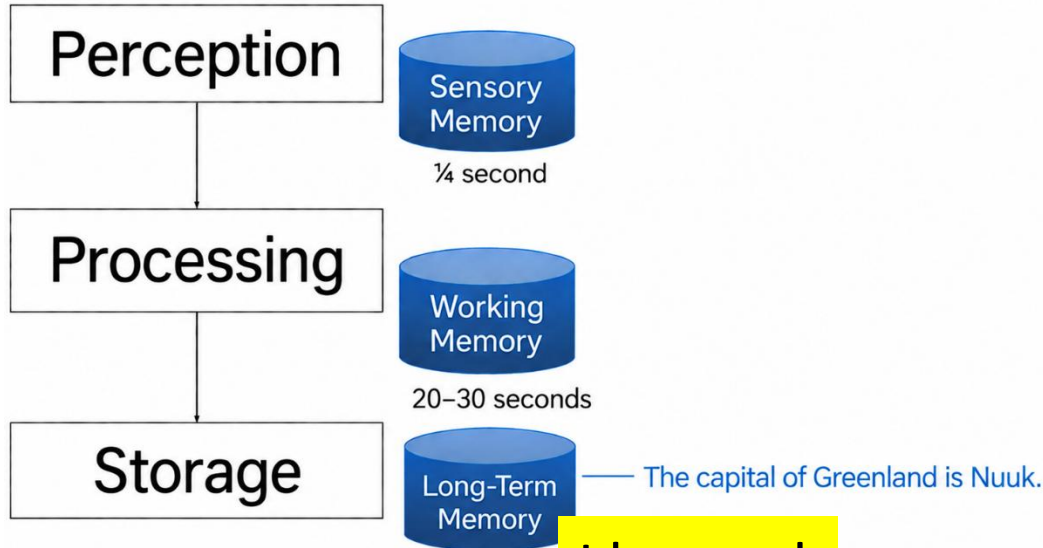
Information



Knowledge

An information processing model

Information



Knowledge

I learned

What determines what gets into and comes out of long-term memory?

Activity 1 – Doctor Visit

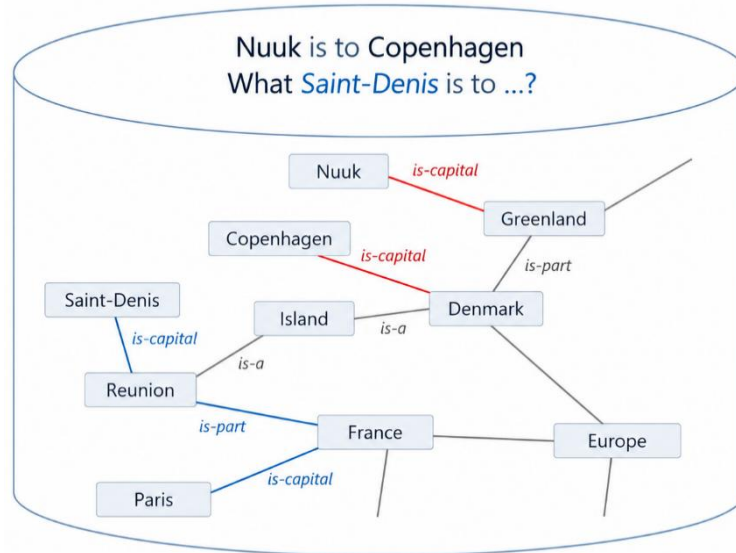
You will see three experiments that investigate people's knowledge and memory for doctor visits. For each experiment, identify patterns you find interesting or surprising. Compare across experiments when useful. Find a way to summarize or visualize the most important patterns. For each pattern, identify the evidence in the data and try to explain why it might occur.

Knowledge organization in long-term memory



Fact: Nuuk is the capital of Greenland.

Schema: Organized structure of knowledge about a concept or situation.



Knowledge organization in long-term memory



Fact: Nuuk is the capital of Greenland.

Schema: Organized structure of knowledge about a concept or situation.

Script: A schema for a familiar sequence of events.

Check in → Wait → Nurse → Doctor → Treatment → Leave

Stereotypical Recall

Check in → **Sit down** → **Read magazine** → Name called
→ Enter exam room → **Nurse tests** → **Doctor examines**
→ Make another appointment → **Leave office**

People remember events of high stereotypy.

Check in → Read magazines → Name called → Undress
→ Talk to nurse → Doctor enters → Doctor friendly →
Prescribed pills → Leave

Script Intrusion

Check in → **Sit down** → Read magazines → Name called
→ Undress → Talk to nurse → **Nurse tests** → Doctor
enters → Doctor friendly → **Doctor examines** →
Prescribed pills → Leave

People falsely remember events from own script.

Check in with receptionist > Four chairs in waiting room > Read magazines > Nurse calls his name > Undress > Put clothes on coat hanger > Talk to nurse > Doctor enters > Doctor is very friendly > Doctor prescribes pills > Forgot wallet > Leave office

P15: Check in > Wait/read > Nurse > Doctor > Prescription > Wallet

P18: Receptionist > Wait > Nurse > Doctor > Wallet > Leave

P20: Check in > Wait/read < Nurse > Doctor > Wallet

Ordered Recall

① Check in with receptionist > ② Four chairs in waiting room > ③ Read magazines > ④ Nurse calls his name > ⑤ Undress > ⑥ Put clothes on coat hanger > ⑦ Talk to nurse > ⑧ Doctor enters > ⑨ Doctor is very friendly > ⑩ Doctor prescribes pills > ⑪ Forgot wallet > ⑫ Leave office

P15: ① Check in ③ Wait/read ④ Nurse ⑦ Doctor ⑩ Prescription ⑪ Wallet

P18: ① Receptionist ③ Wait ④ Nurse ⑦ Doctor ⑪ Wallet ⑫ Leave

P20: ① Check in ③ Wait/read ④ Nurse ⑦ Doctor ⑪ Wallet

People remember events in chronological sequence.

Stereotypical Recall

Check in with receptionist > Four chairs in waiting room > Read magazines > Nurse calls his name > Undress > Put clothes on coat hanger > Talk to nurse > Doctor enters > Doctor is very friendly > Doctor prescribes pills > Forgot wallet > Leave office

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P18: Receptionist > Wait > Nurse > Doctor > Wallet > Leave

P20: Check in > Wait/read < Nurse > Doctor > Wallet

People remember events of high stereotypy.

Obstacle Recall

Check in with receptionist > Four chairs in waiting room > Read magazines > Nurse calls his name > Undress > Put clothes on coat hanger > Talk to nurse > Doctor enters > Doctor is very friendly > Doctor prescribes pills > **Forgot wallet** > Leave office

P15: Check in > Wait/read > Nurse > Doctor > Prescription > **Wallet**

P18: Receptionist > Wait > Nurse > Doctor > **Wallet** > Leave

P20: Check in > Wait/read < Nurse > Doctor > **Wallet**

People remember obstacles to goal completion.

Activity 2 – Balloon Passage

You will see two experiments investigating people's memory for a passage. For each experiment, identify patterns you find interesting or surprising. Compare across experiments when useful. Find a way to summarize or visualize the most important patterns. For each pattern, identify the evidence in the data and try to explain why it might occur.

1 — If the balloons popped, the sound would not be able to carry.

2 — Everything would be too far away from the correct floor.

...

12 — Then there would be fewer potential problems.

13 — With face to face contact, the least number of things could go wrong.

Immediate Recall

P1: 13 – 12 – 1 – 8 – 3

P2: 13 – 11 – 1 – 5

P3: 13 – 1 – 9 – 4

P4: 12 – 13 – 1 – 10 – 6

Delayed Recall

P5: 1 – 2 – 5

P6: 1 – 7

P7: 3 – 1 – 13 – 8

P8: 1 – 2 – 10

Primacy and Recency

- 1** — If the balloons popped, the sound would not be able to carry.
- 2** — Everything would be too far away from the correct floor.
- ...
- 12** — Then there would be fewer potential problems.
- 13** — With face to face contact, the least number of things could go wrong.

Immediate Recall

P1: 13 – 12 – 1 – 8 – 3

P2: 13 – 11 – 1 – 5

P3: 13 – 1 – 9 – 4

P4: 12 – 13 – 1 – 10 – 6

Delayed Recall

P5: 1 – 2 – 5

P6: 1 – 7

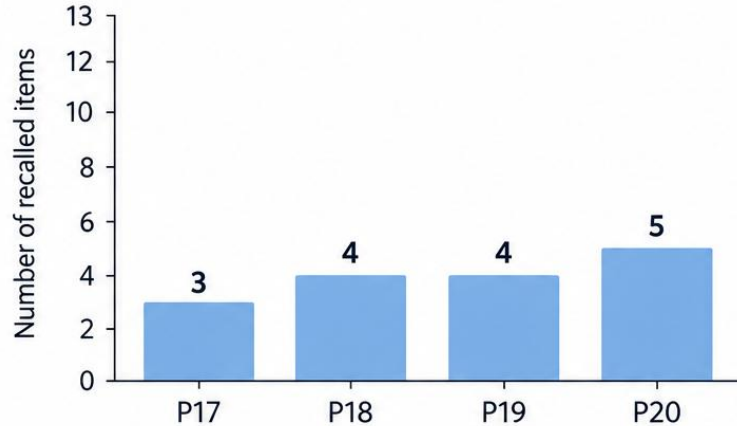
P7: 3 – 1 – 13 – 8

P8: 1 – 2 – 10

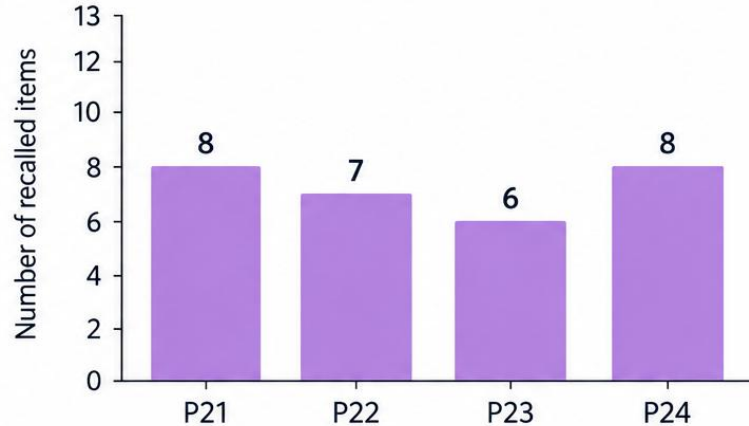
People remember first and last items of a stimulus list.

Total Recall

Double encode
Read passage twice



Context before
See picture → read passage



People remember more if material is meaningful.

Verbatim vs. Gist

Passage: If the balloons popped, the sound would not be able to carry.

P17: If the balloons popped, the sound would not be able to carry.

P21: If the balloons pop, the sound won't reach the window.

Meaning leads to gist recall; nonsense leads to verbatim recall.

Inference intrusions

Passage: The human voice is not loud enough to carry that far.

P23: But his voice won't be loud enough **to reach the window.**

Passage: A break in the middle of the wire would also cause problems.

P21: He also needs a good flow of electricity **to the speaker.**

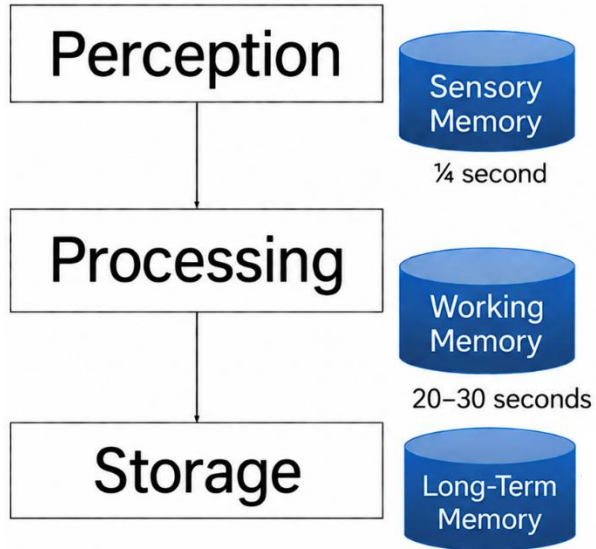
People remember inferences made while reading a passage.

Memory Concepts

Concept Type	What we learned
Schema concepts	
Stereotypical recall	People remember events of high stereotypy.
Script intrusions	People falsely remember events from their own script.
Ordered recall	People remember events in chronological sequence.
Obstacle recall	People remember obstacles to goal completion.
Encoding concepts	
Total recall	People remember more if material is meaningful.
Primacy and recency	People remember first and last items of a stimulus list.
Gist and verbatim	Meaning leads to gist recall; nonsense leads to verbatim recall.
Inference intrusions	People remember inferences made while reading a passage.

Implications for learning

Information

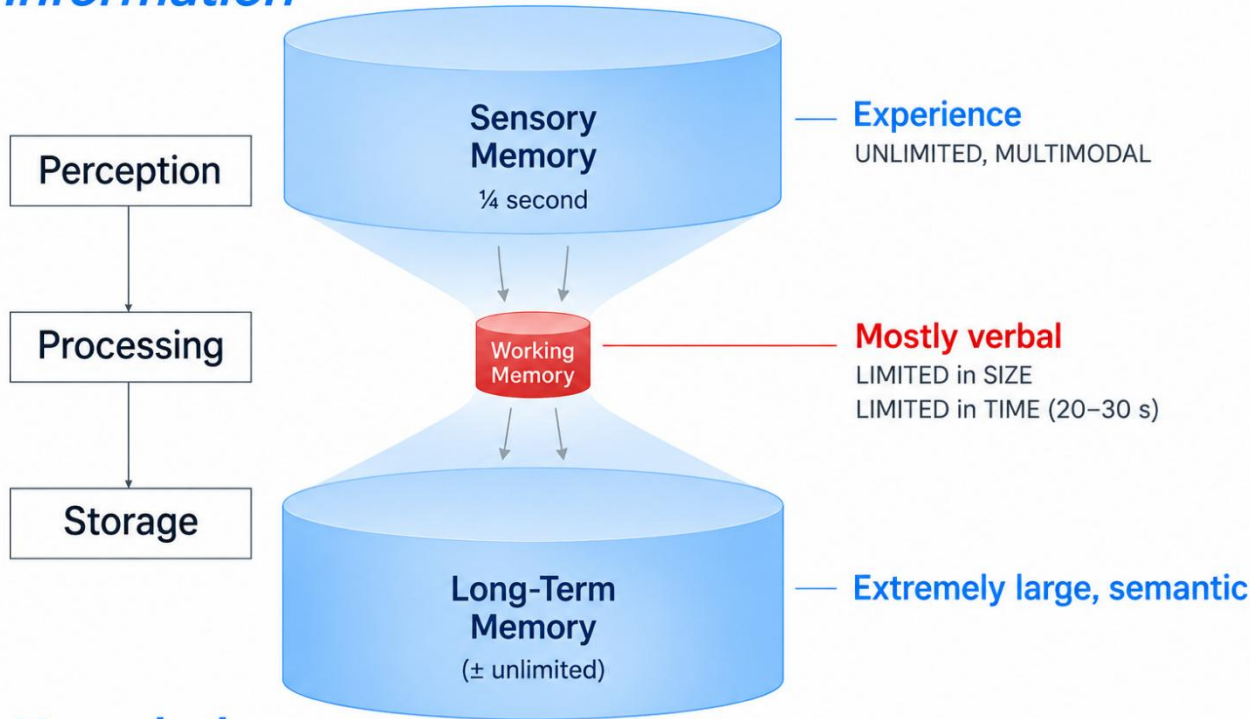


Knowledge

- Learning comes from processing information perceived or stored in long-term memory
- Processing information is managed by the working memory

Working memory is limited

Information



Knowledge

1. Nathalie is the sister of Suzanne
2. Suzanne is the daughter of Lucie
3. Natasha is the daughter of Nathalie
4. Gloria is the sister of Natasha

Who is the grandmother of
Gloria?

1. Luc is the father of Mike
2. Mike is brother of Emile
3. John is the son of Emile
4. Nicolas is the brother of John
5. Julien is the son of Mike

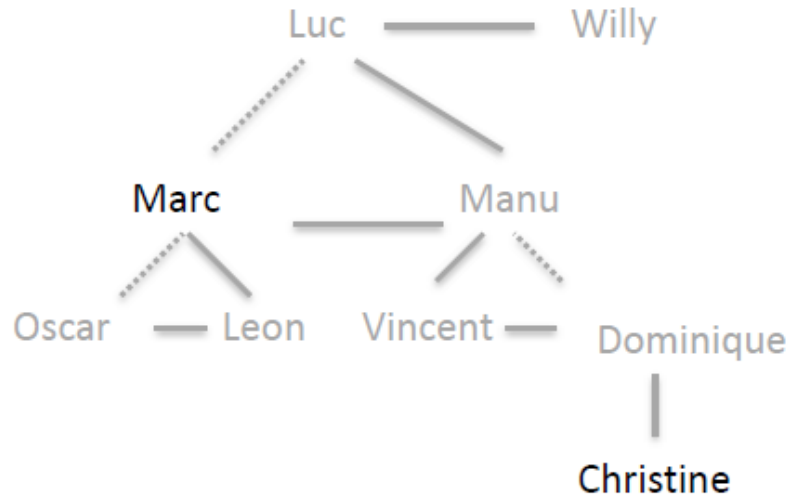
Who is the cousin of Nicolas?

1. Olga is the sister of Michelle
2. Michelle is the daughter of Lucie
3. Natasha is the daughter of Olga
4. Camilla is the sister of Natasha
5. Paul is the husband of Camilla
6. Donald is the son of Paul

Who is the grandmother of Donald?

1. Marc and Manu are brothers
2. Leon and Oscar are brothers
3. Leon is the son of Marc
4. Vincent is the son of Manu
5. Willy is the uncle of Marc
6. Dominique is the brother of Vincent
7. Christine is the daughter of Dominique

Who is the brother of Christine's grandfather?



Cognitive Overload

Increasing demand →

Michel est le père de
Martine

Martine est la sœur
de Serge

Serge est le père de
Tristan

Marc est le frère de
Michel

Michel est le père de
Martine

Martine est la sœur de
Serge

Serge est le père de
Tristan

Luc est le frère de Tristan

Lena est la sœur de Lucie

Marc est le frère de Michel

Michel est le père de
Martine

M

iv.

Martine est la sœur de
Serge

Serge est le père de Tristan

Luc est le frère de Tristan

Lucie est la mère de Serge

Sophie est la sœur de
Serge

Lena est la sœur de Lucie

Marc est le frère de Michel

Michel est le père de
Martine

Martine est la sœur de
Serge

Serge est le père de Tristan

Luc est le frère de Tristan

Lucie est la mère de Serge

Nadine est la fille de Tristan

Paul est le frère de Sophie

Sophie est la sœur de
Serge

Lena est la sœur de Lucie

Marc est le frère de Michel

Michel est le père de
Martine

Martine est la sœur de
Serge

Serge est le père de Tristan

Luc est le frère de Tristan

Lucie est la mère de Serge

Nadine est la fille de
Tristan

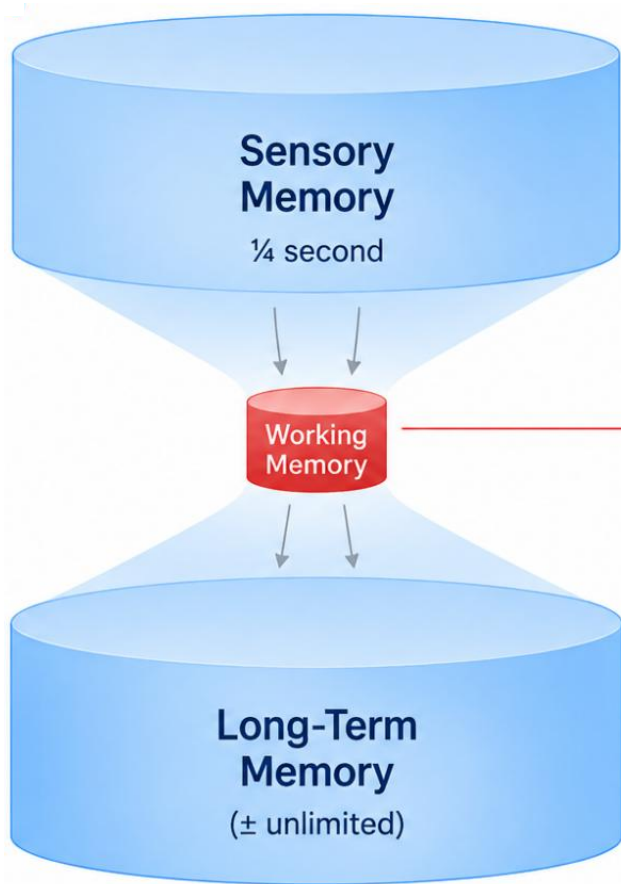
$$5^? \pm 2$$

Trying to process all this information while also doing the additional task
can overload working memory.

Learning comes from processing information perceived or stored in long-term memory

Processing information is managed by the working memory

If **cognitive overload**, processing will fail, hence **no learning**
If no processing, **no cognitive load** but **no learning**



Intrinsic Cognitive Load: some things are harder to learn

Extraneous Cognitive Load: due to bad design

Germane Cognitive Load: the effort to create schemas

Intrinsic



The task itself is
demanding

Extraneous



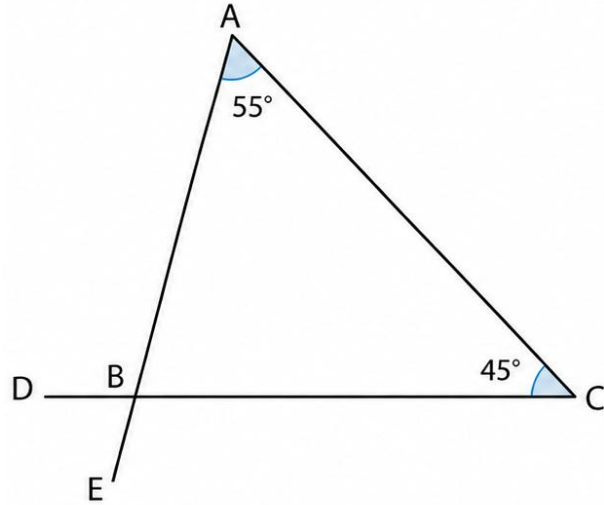
Added difficulty
that is irrelevant
to the goal

Germane



Productive effort that
supports learning and
skill development

Reduce extraneous load

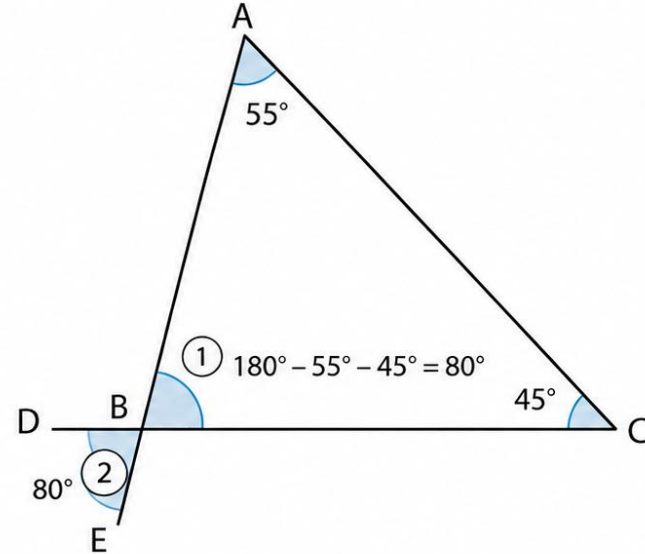


In the above figure, find a value for Angle DBE.

Solution:

$$\begin{aligned}\text{Angle } ABC &= 180^\circ - \text{Angle } BAC - \text{Angle } BCA \\ &\quad (\text{internal angles of a triangle sum to } 180^\circ) \\ &= 180^\circ - 55^\circ - 45^\circ \\ &= 80^\circ\end{aligned}$$

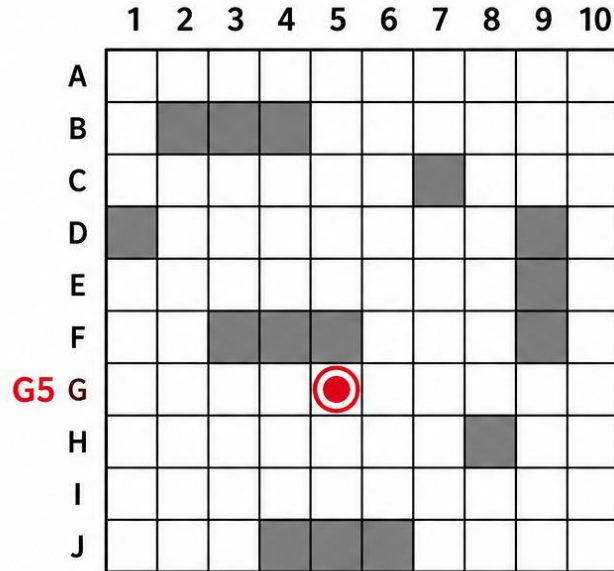
$$\begin{aligned}\text{Angle } DBE &= \text{Angle } ABC \quad (\text{vertically opposite angles are equal}) \\ &= 80^\circ\end{aligned}$$



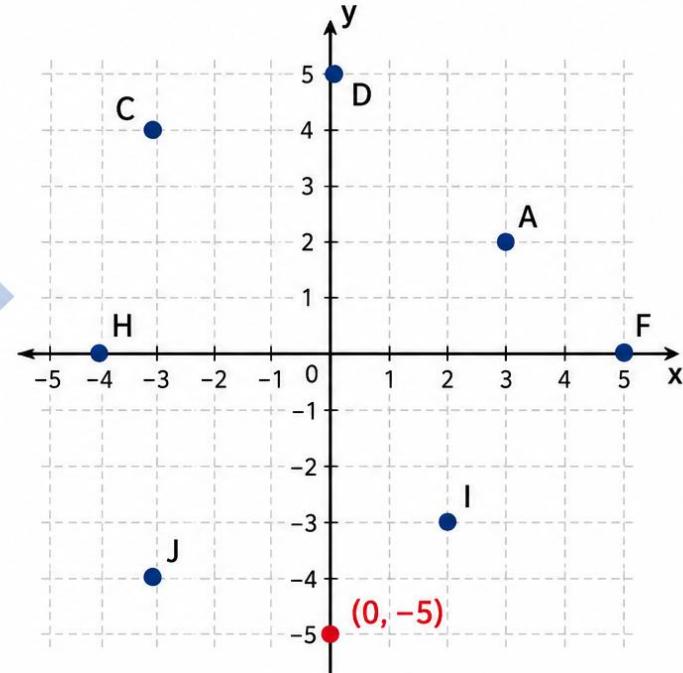
Split Attention Effect

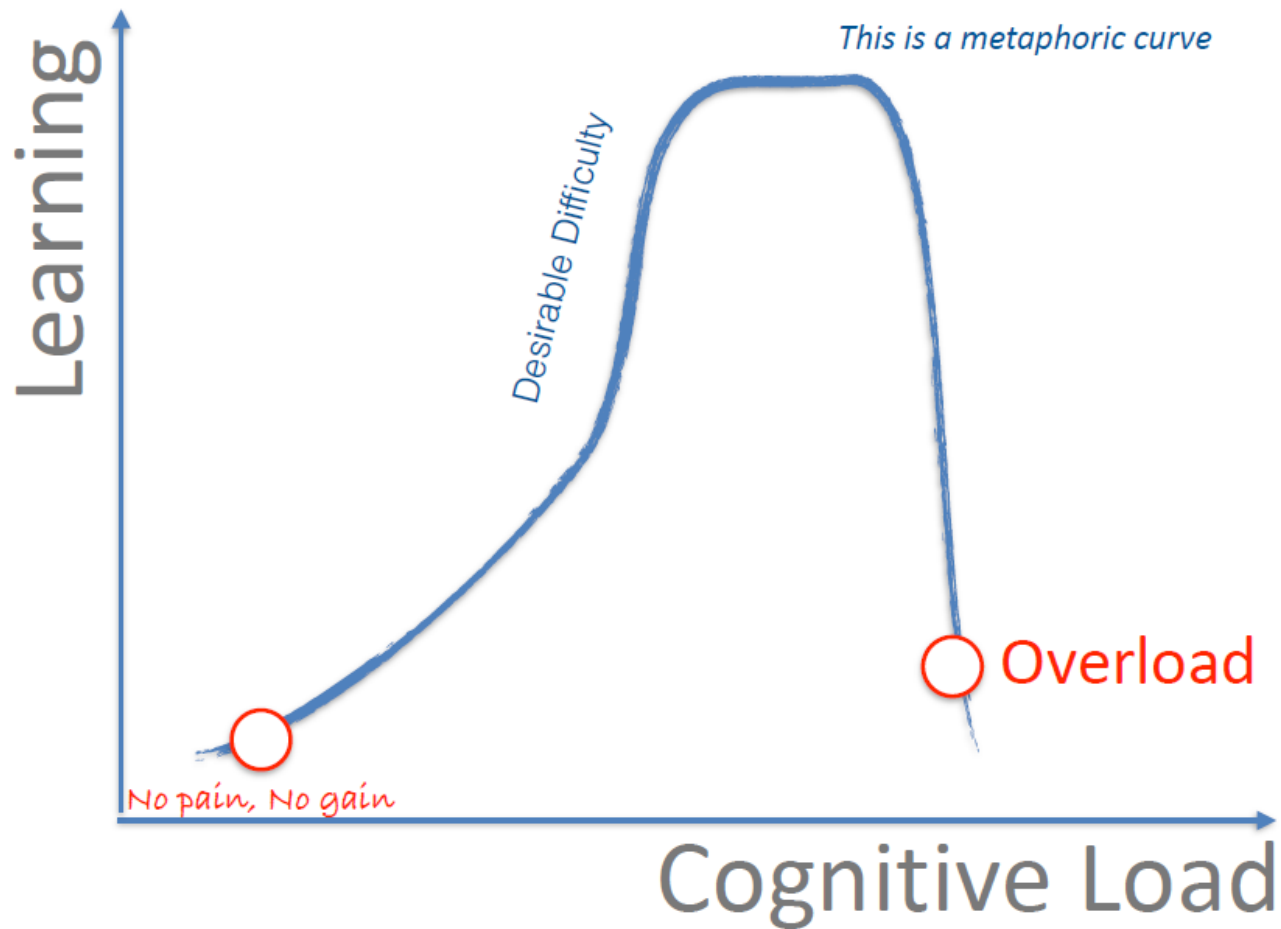
Manage intrinsic load

Battleship



Cartesian Coordinates System





Take-Aways for instruction

- ① **Reduce extraneous load**
Avoid unnecessary processing
- ② **Manage intrinsic load**
Make necessary complexity manageable
- ③ **Create productive effort**
Engage learners in processing that builds knowledge and skills